Dynamics of Self-propelled particles in laminar flow

Under the Guidance of:
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Department of Physics

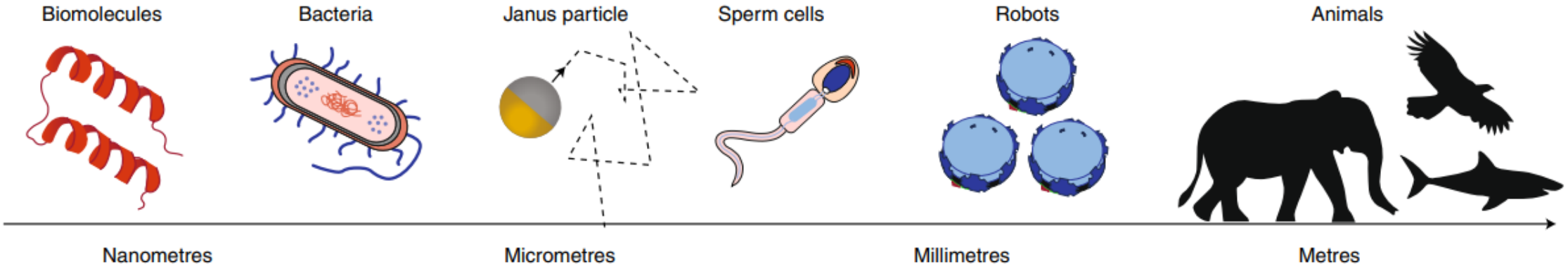
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Motivation and Background

- Active particles are self-propelled entities interacting with fluid flows.
- These particles don’t just drift randomly and aimlessly, instead they have unique ways of moving and can even respond to their surroundings.
- Understanding their dynamics is crucial for biological and technological applications.



Active matter systems across scales: From nanoscale biomolecules to microscale particles like bacteria and Janus particles, to millimeter-scale robots, and up to meter-scale animals and organisms, demonstrating the universality of active matter dynamics.

Image Credit : Cichos, F., Gustavsson, K., Mehlig, B. *et al.* Machine learning for active matter.

Active Particle Dynamics

Rate of change of particles' position over time

$$\dot{\vec{r}} = \mathbf{v_f}(\mathbf{r}, t) + v_p \hat{\mathbf{p}},$$

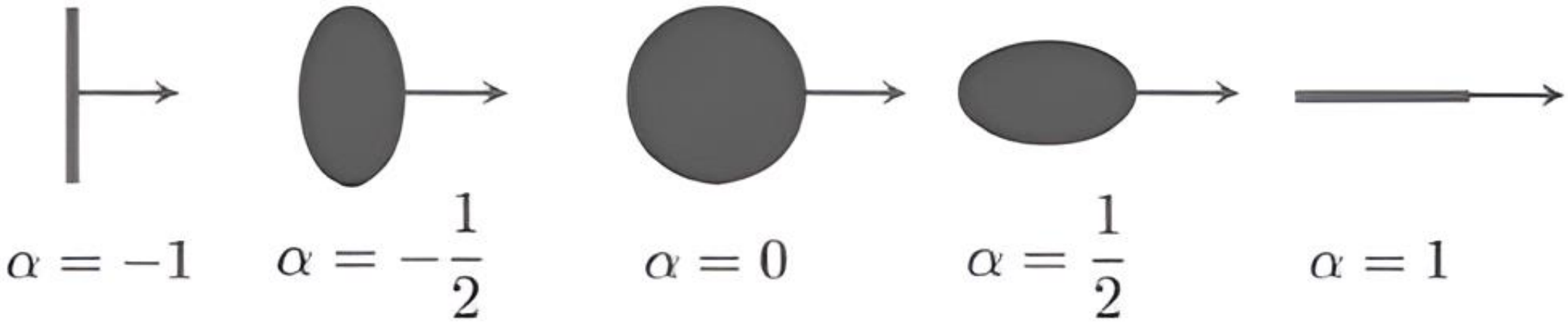
Velocity field of the fluid

Self-propulsion velocity of the particle

Particle's Orientation

The **alignment parameter** α characterizes the shape of the ellipse . For a sphere , γ = 1

$$\alpha = \frac{\gamma^2 - 1}{\gamma^2 + 1},$$



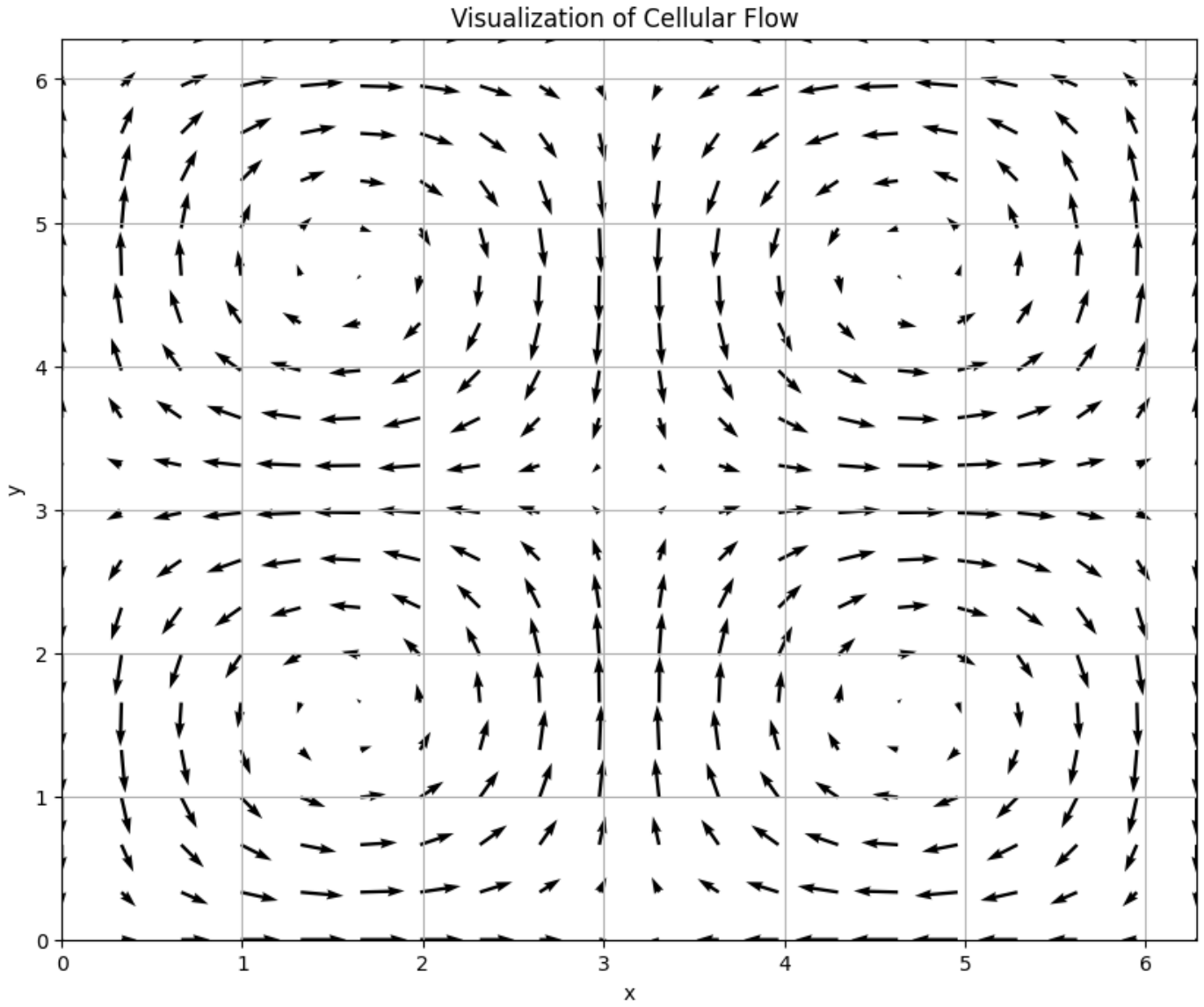
Streamline Function

$$\psi(x,y,t) = \frac{U_0 L}{2\pi} \sin\left(\frac{2\pi(x - x_0(t))}{L}\right) \sin\left(\frac{2\pi y}{L}\right),$$

Simplifies to : $$\psi(x,y) = \sin(x)\sin(y),$$

$$u_x = \frac{\partial \psi}{\partial y} \quad \text{and} \quad u_y = -\frac{\partial \psi}{\partial x}.$$

Vector field: $$\vec{v} = \sin(x)\cos(y)\,\hat{i} - \cos(x)\sin(y)\,\hat{j}$$



Cellular flow field used in this study

The orientation of ellipsoidal particles in a fluid is governed by :

$$\dot{\mathbf{p}} = \tfrac{1}{2}\boldsymbol{\omega} \times \mathbf{p} + \alpha \mathbf{p} \cdot \mathbf{E} \cdot (\mathbf{I} - \mathbf{pp}),$$

In two dimensions the orientation of the self-propelled particles can be represented by an angle in the *x-y* plane :

$$\dot{\theta} = \alpha \left[\frac{1}{2} \left(\frac{\partial^2 \psi}{\partial y^2} - \frac{\partial^2 \psi}{\partial x^2} \right) \cos 2\theta - \frac{\partial^2 \psi}{\partial x \partial y} \sin 2\theta \right] - \frac{1}{2} \left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} \right) .$$

Jeffrey's Equation (G. B. Jeffery, Proc. R. Soc. A **102**, 161 (1922))

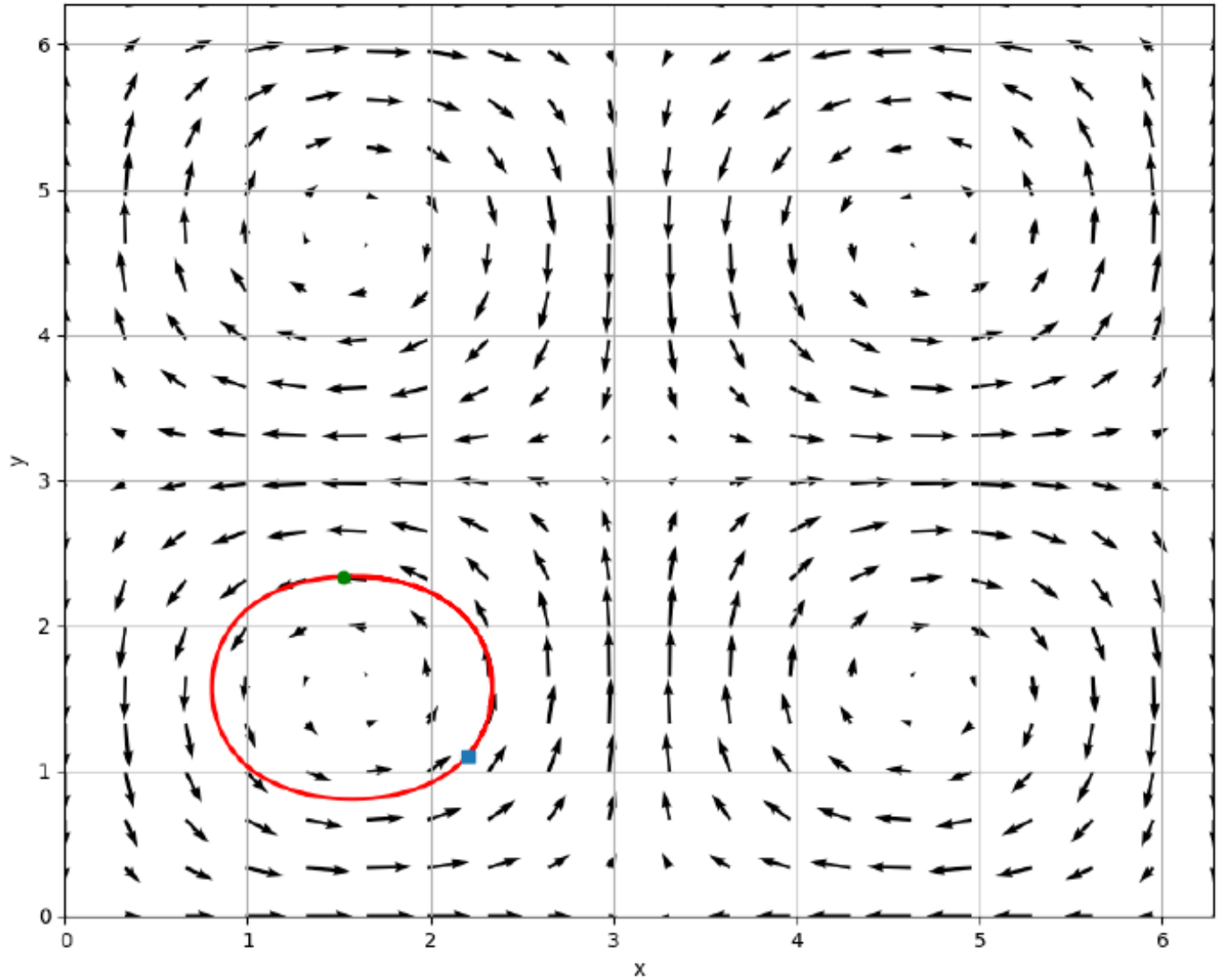
The orientation dynamics are given by :

$$\dot{\theta} = \sin(x)\sin(y)$$

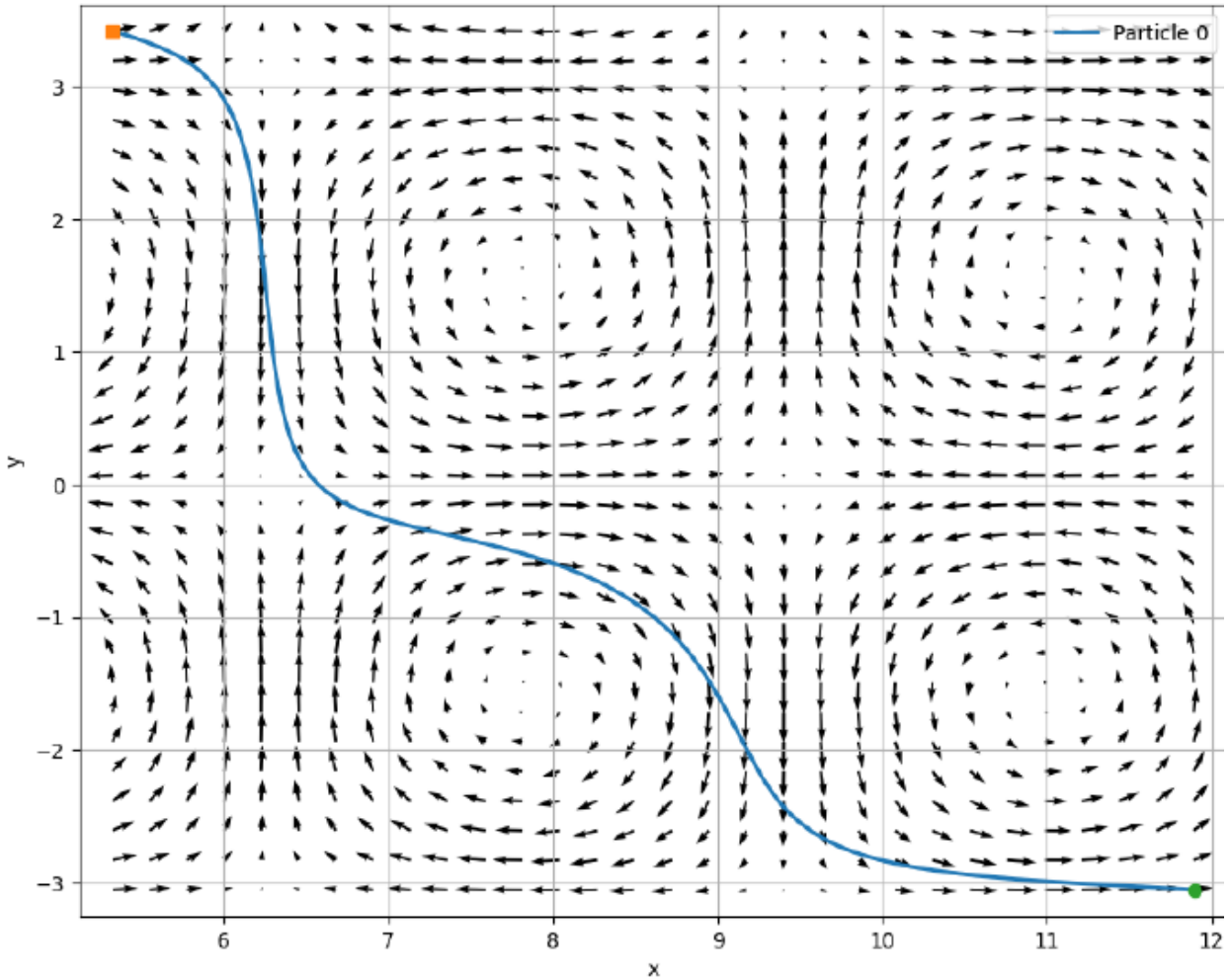
The translational motion is described by :

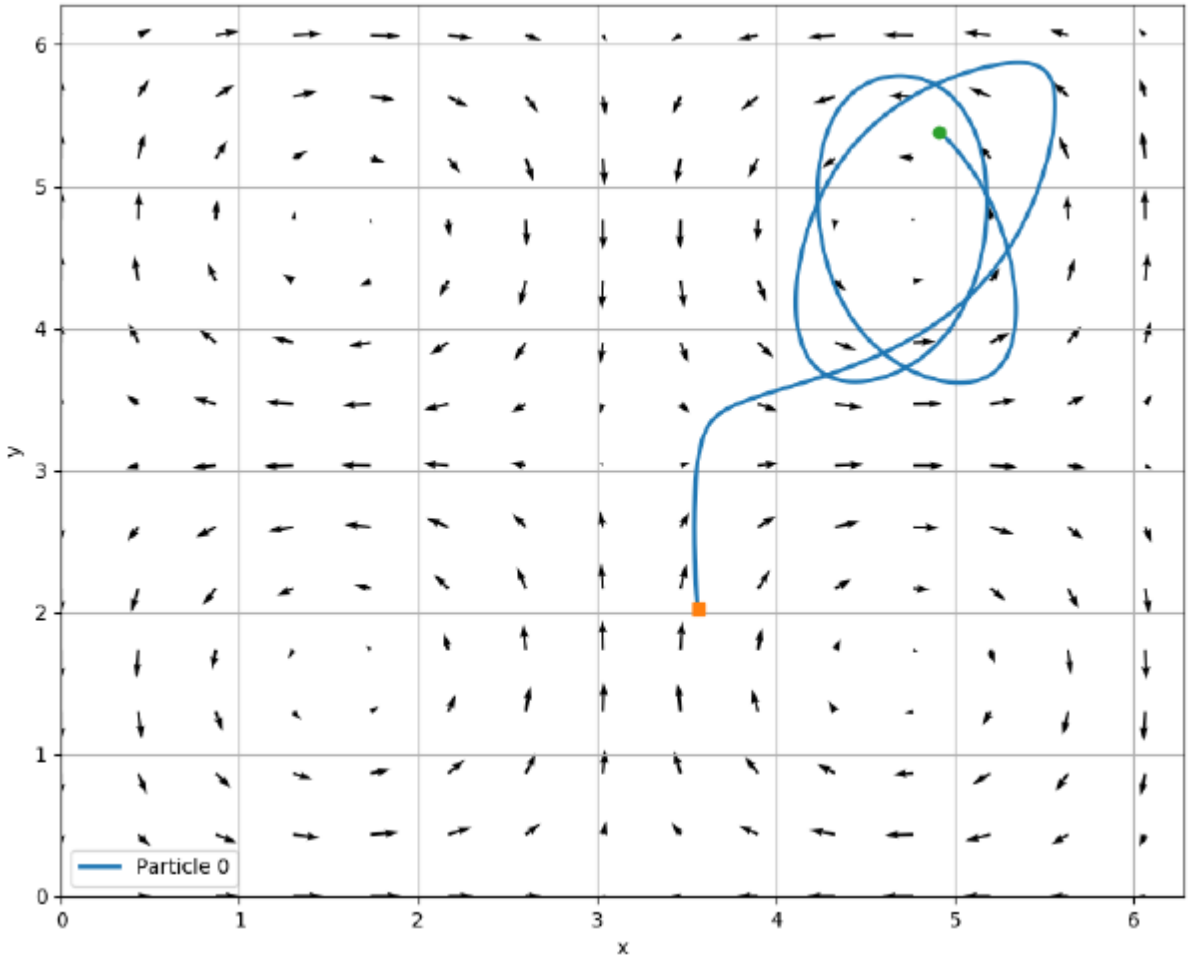
$$\dot{x} = \sin(x)\cos(y) + v_s\cos(\theta)$$

$$\dot{y} = -\cos(x)\sin(y) + v_s\sin(\theta)$$



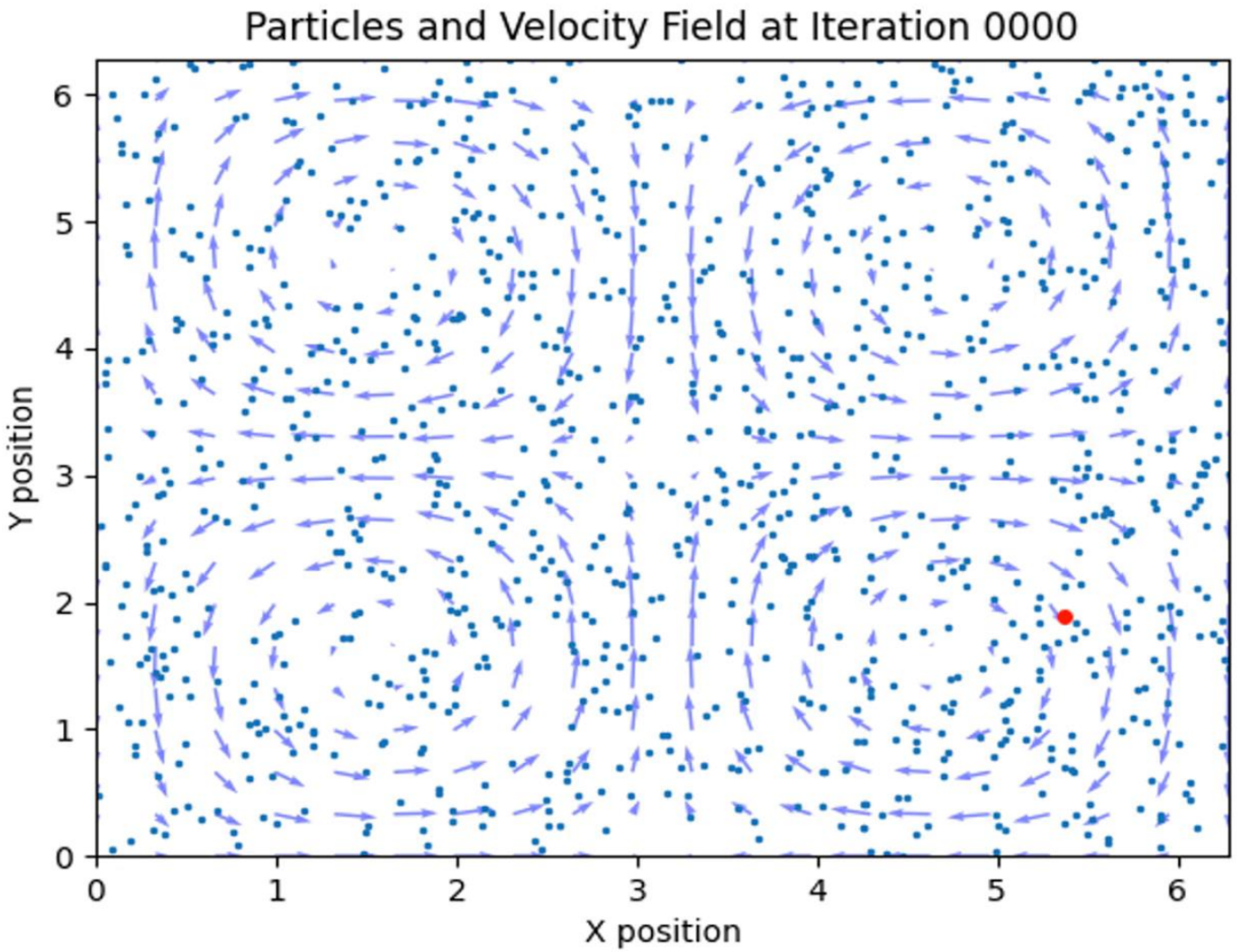
Motion of a tracer (v_s=0) in cellular flow



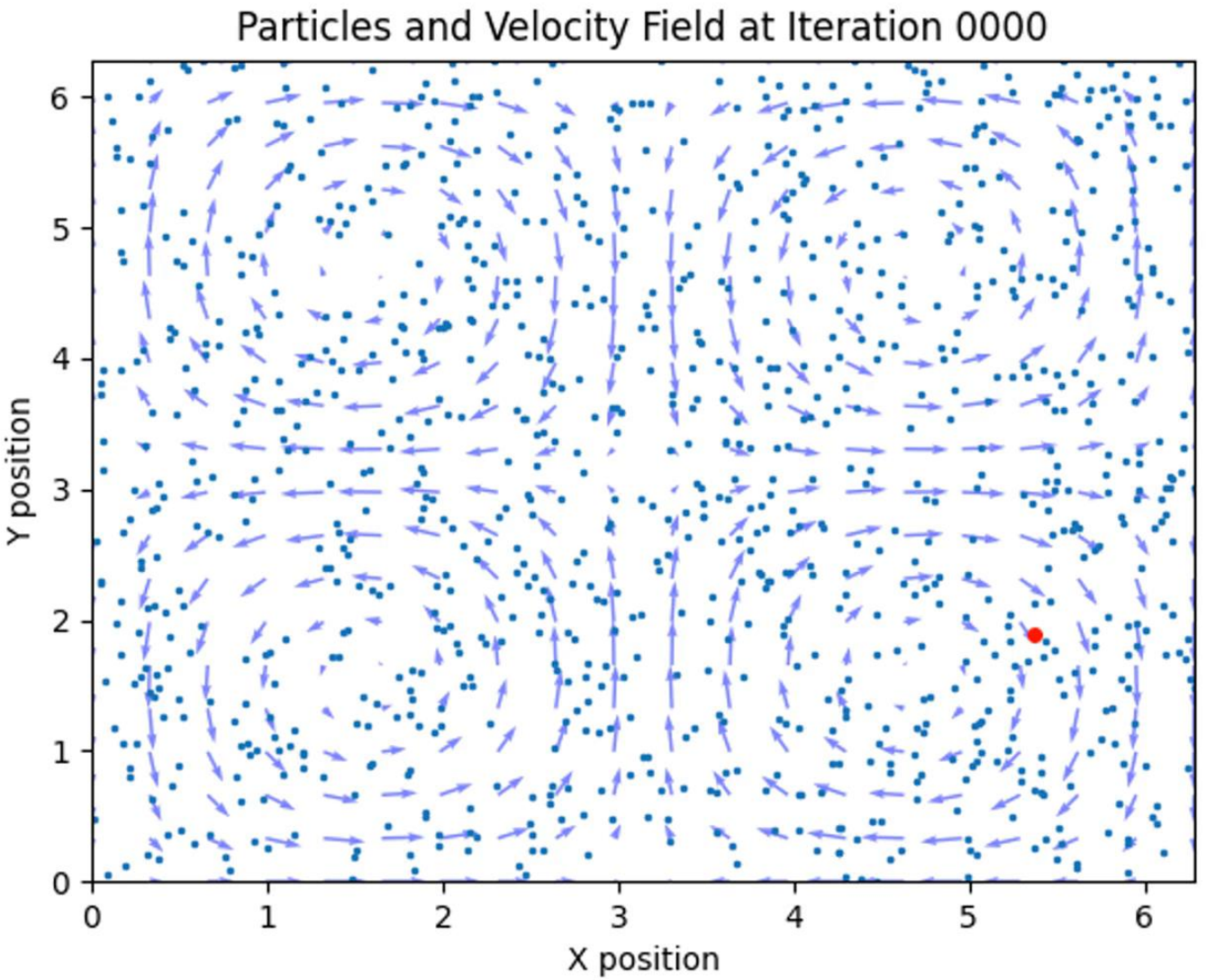


Motion of an active particle(v_s=0.5) in cellular flow.

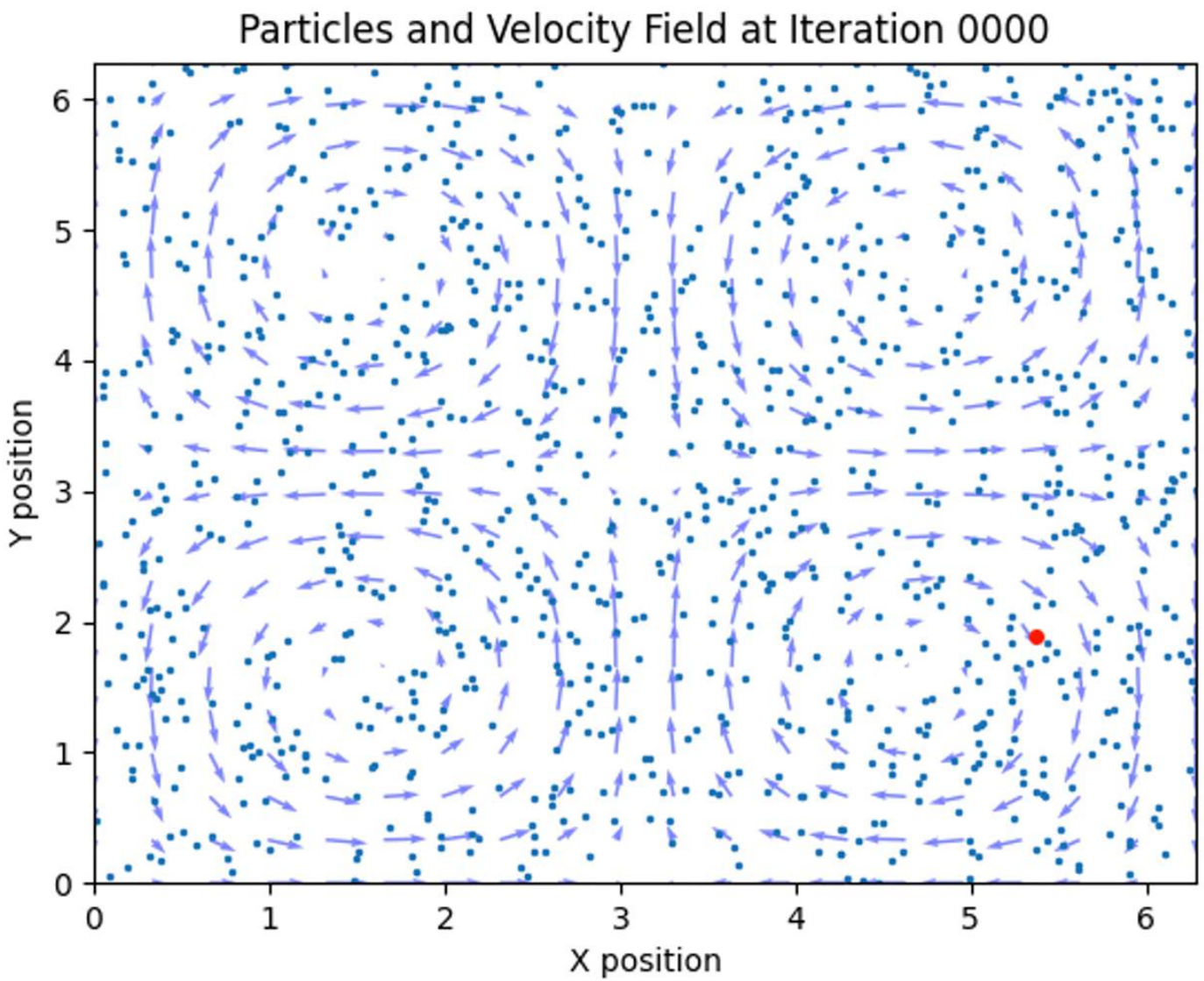
Swimming velocity $\mathbf{v_s = 0.0}$



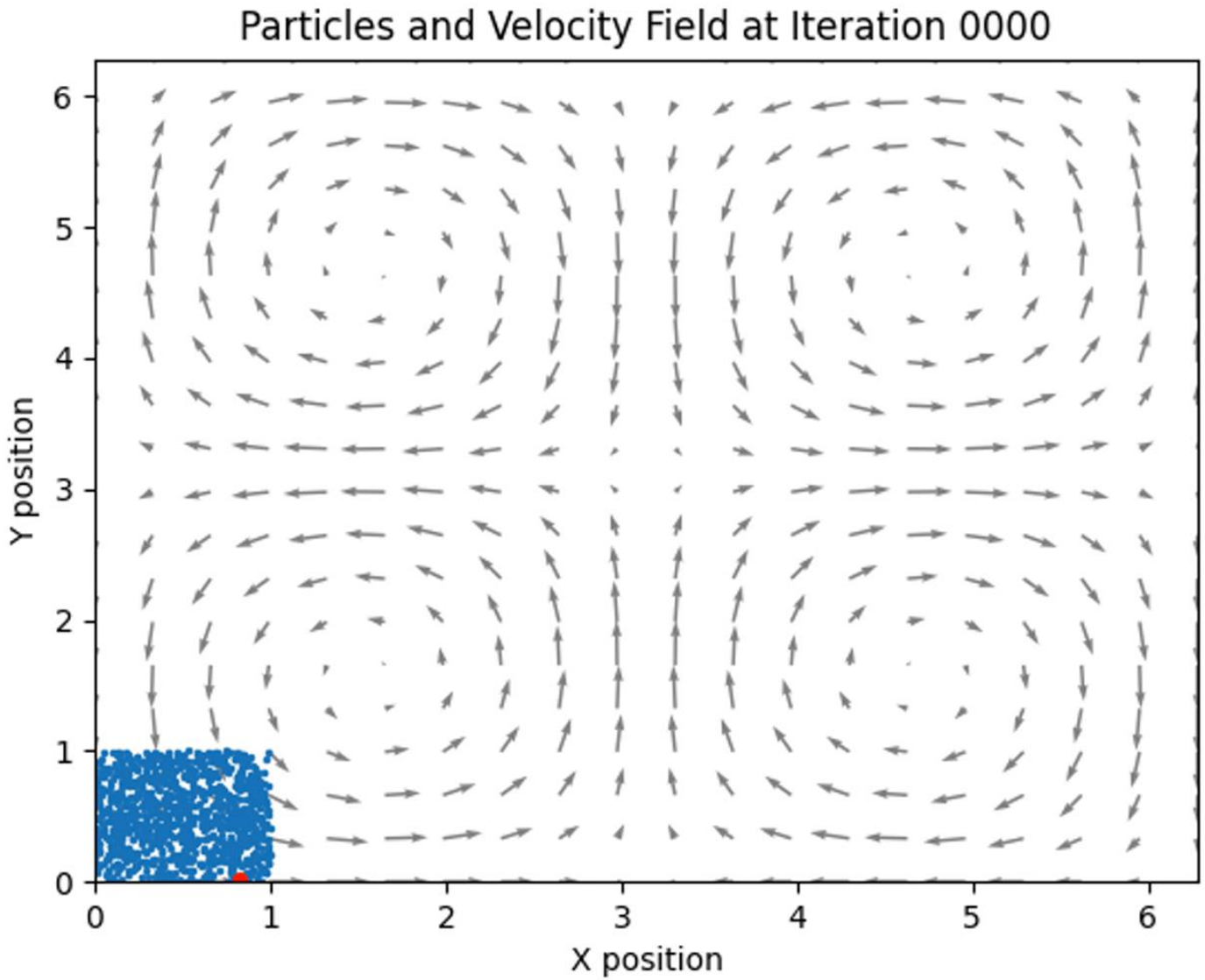
Swimming velocity $\mathbf{v_s = 1.0}$



Swimming velocity $\mathbf{v_s = 5.0}$



Swimming velocity $v_s = 0.5$



Mean Squared Displacement (MSD)

MSD measures how far particles spread out from their starting(reference) point.

Defined as : $\mathrm{MSD}(t) = \langle |\mathbf{r}(t) - \mathbf{r}(0)|^2 \rangle,$

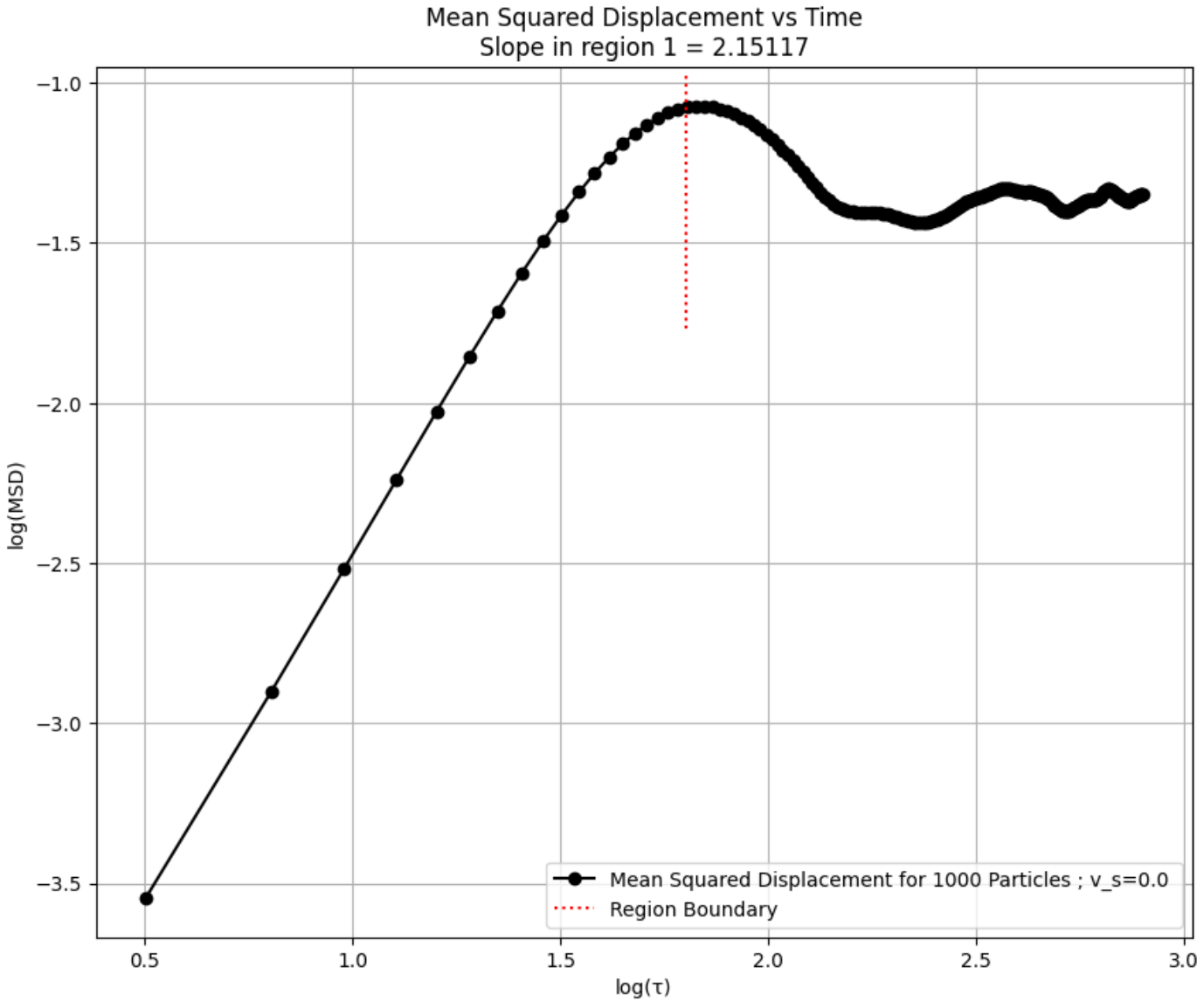
Follows a power law : $\mathrm{MSD}(t) \propto t^{\beta},$

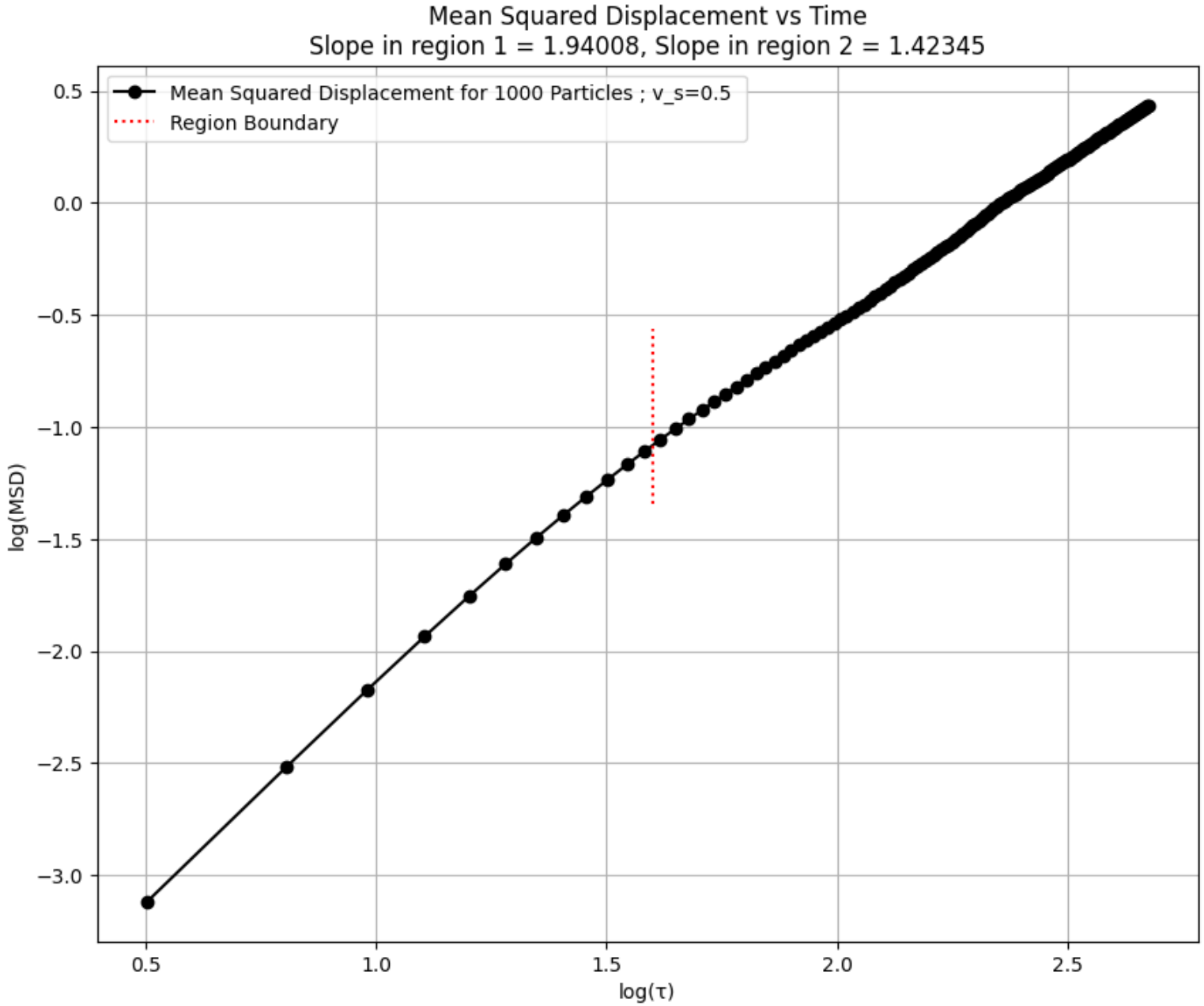
$$\log(\mathrm{MSD}(t)) = \beta \cdot \log(t) + \text{constant}.$$

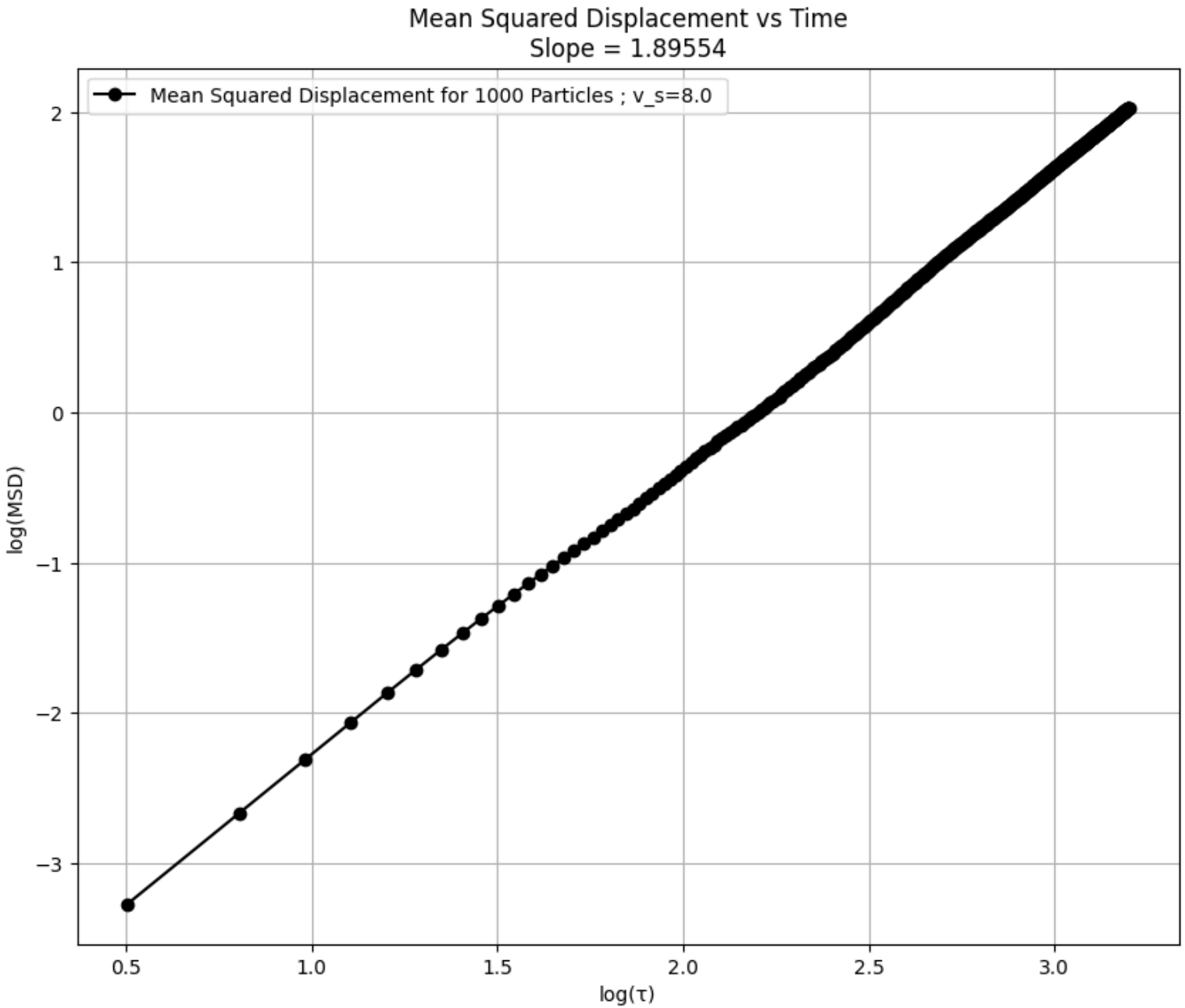
Exponent (ß)	**Type of Motion**	**Characteristics**
ß=1	Normal Diffusion	Particles spread out linearly over time (Brownian Motion)
0<ß<1	Sub-diffusion	Particles move slowly due to obstacles or constraints
ß=2	Ballistic	Particles move in straight lines with constant velocity

log(MSD) vs log(t) graph

- Two Regimes Observed: Initial ballistic-like growth transitions to a saturation phase.
- Short-Time Behavior: Linear growth in MSD, indicating unconfined motion.
- Long-Time Behavior: MSD slope approaches zero, reflecting particle confinement in flow cells.

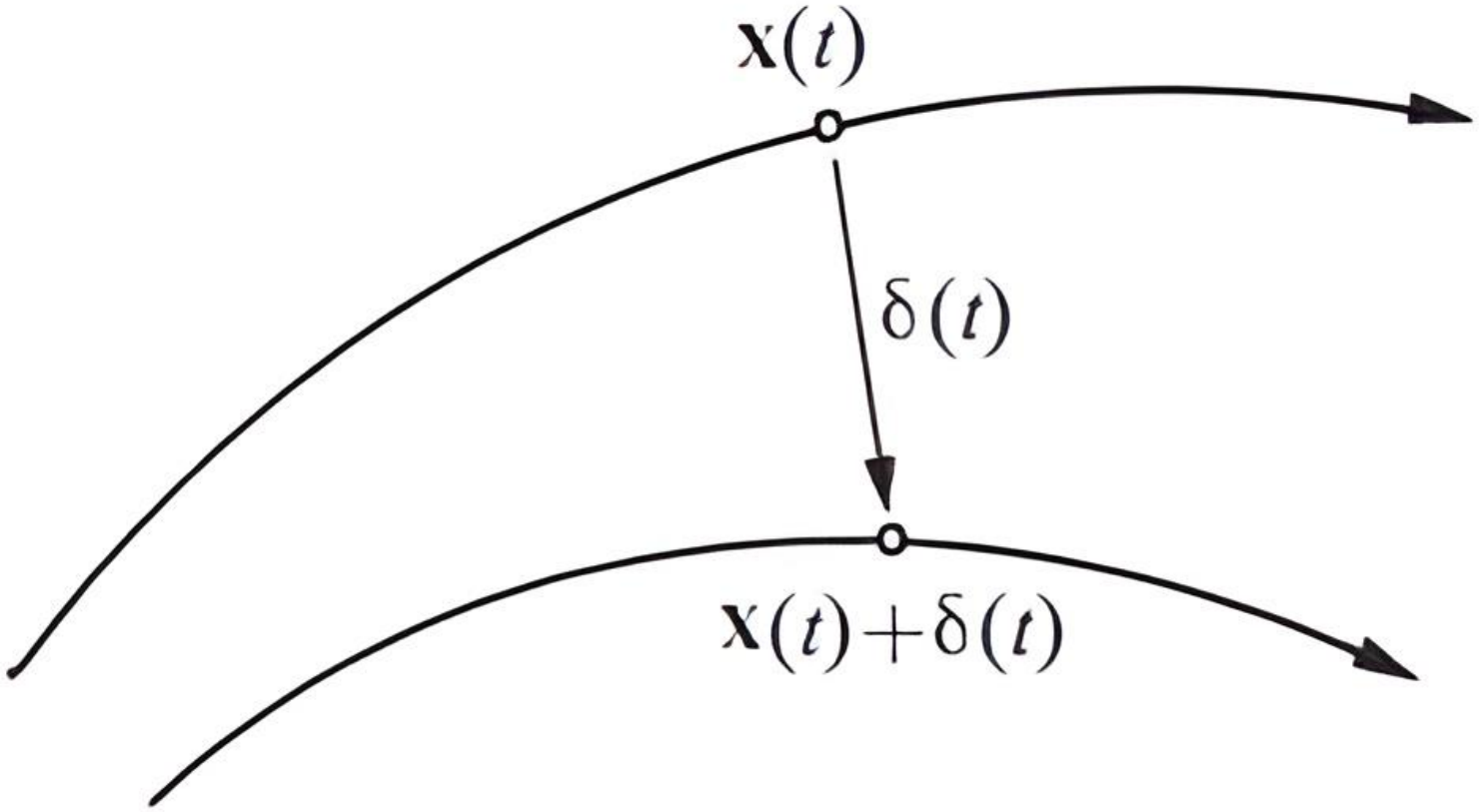



Mean Squared Displacement vs Time
Slope in region 1 = 1.94008, Slope in region 2 = 1.42345
Mean Squared Displacement for 1000 Particles ; v_s=0.5
Region Boundary
log(MSD)
log(τ)
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
0.5
1.0
1.5
2.0
2.5


Mean Squared Displacement vs Time
Slope = 1.89554
Mean Squared Displacement for 1000 Particles ; v_s=8.0
log(MSD)
log(τ)
2
1
0
−1
−2
−3
0.5
1.0
1.5
2.0
2.5
3.0

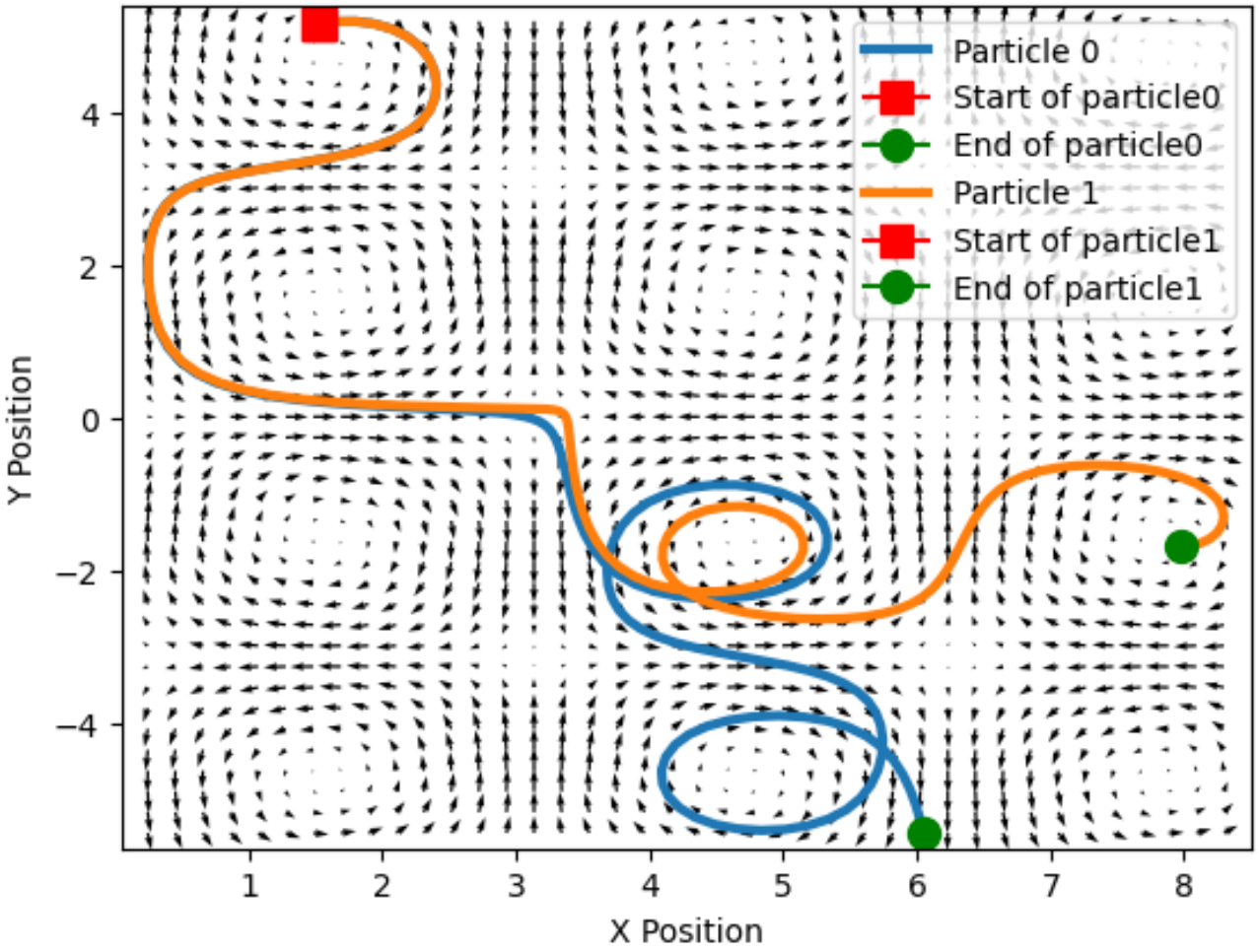
Chaos

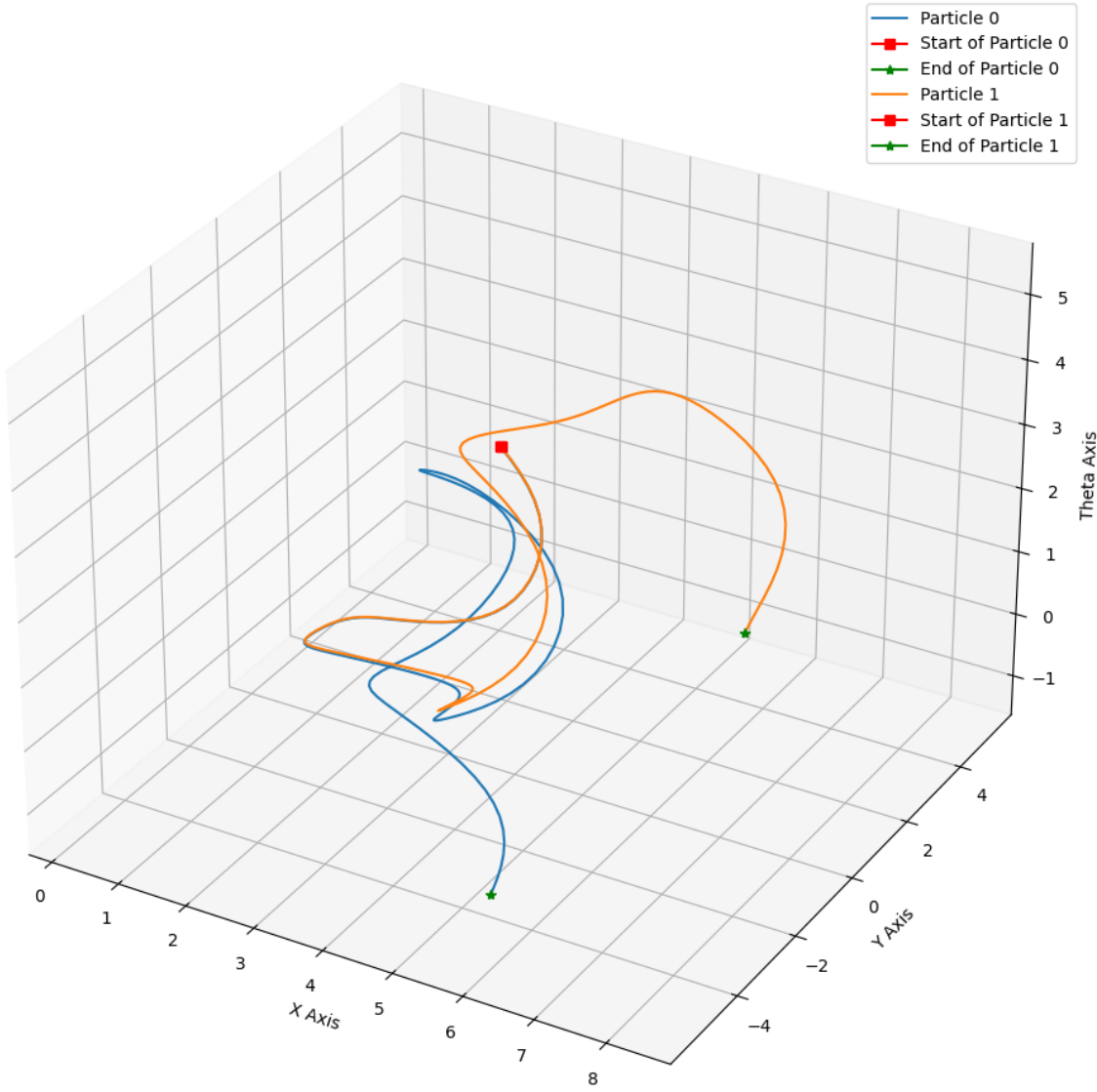
- Chaos theory studies systems that exhibit a high **sensitivity to initial conditions**.
- Even a small perturbation can lead to significantly different outcomes over time.
- The nonlinear terms in the equations (sine and cosine functions) create complex trajectories that depend sensitively on the initial state.



Evolution of a trajectory x(t) and a perturbed trajectory x(t) + δ(t), with δ(t) indicating the deviation between them.

Image Credit : Nonlinear Dynamics and Chaos Steven_H._Strogatz





Comparison of particle trajectories in 2D and 3D phase spaces

Quantifying Chaos through Lyapunov Exponents

- To quantify chaos in your system, we calculate the **Lyapunov exponent**, which is a measure of how quickly two initially close trajectories diverge over time.
- A **positive Lyapunov exponent** indicates that small initial differences grow exponentially, leading to chaotic behavior whereas **Zero or Negative Lyapunov Exponent** indicates no exponential divergence(stable dynamics).

Let’s Calculate the Lyapunov Exponent

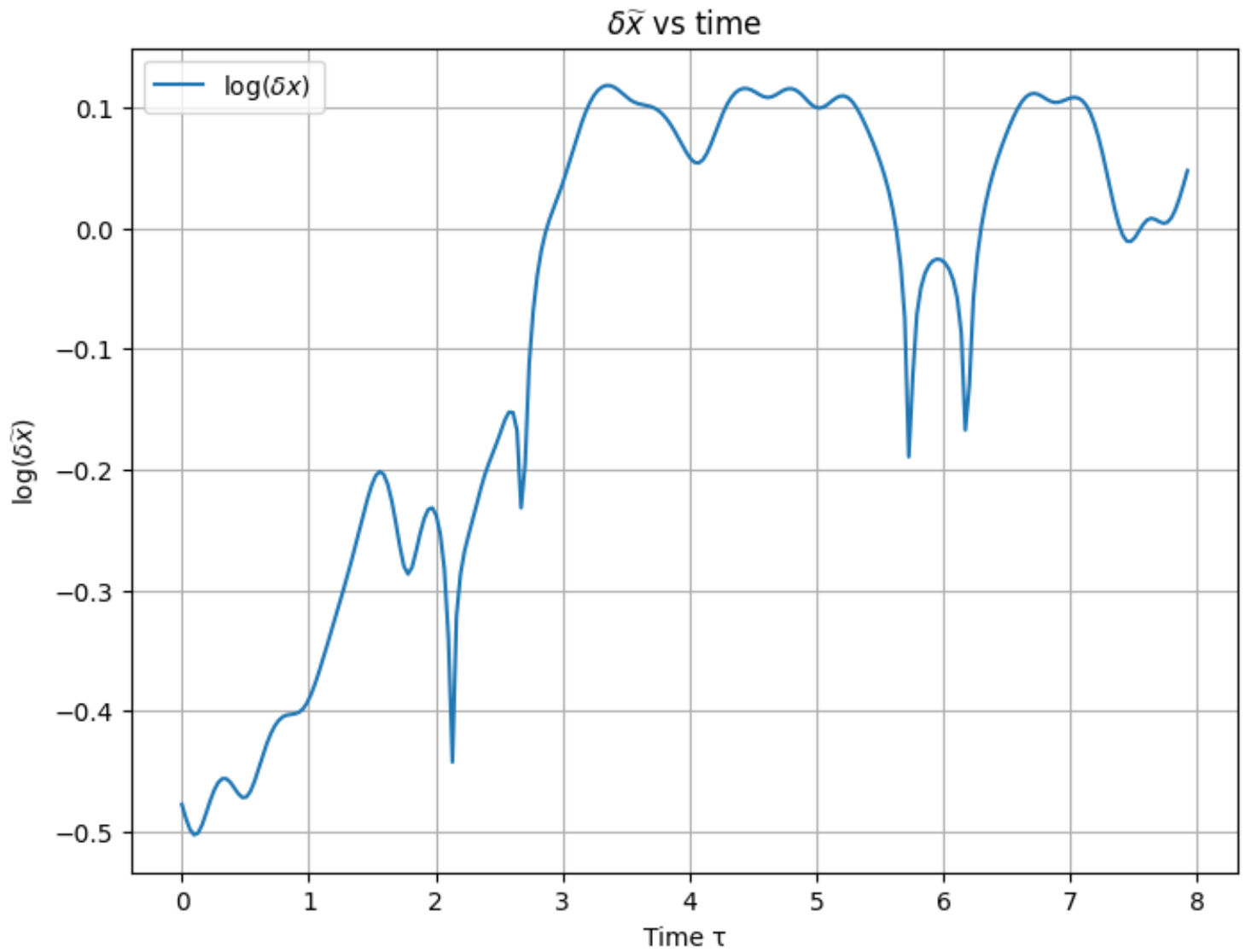
- If two particles start with an initial separation δx(0),their separation at a later time t is given by:

$$\delta x(t) = \delta x(0) e^{\lambda t}$$

- δx(0): Initial separation between two trajectories.
- λ: Lyapunov exponent (what we’re calculating)
- t: Time.

- Taking the natural logarithm:

$$\ln(\delta x(t)) = \ln(\delta x(0)) + \lambda t$$


$\delta\tilde{x}$ vs time
log(δx)
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
log($\delta\tilde{x}$)
0
1
2
3
4
5
6
7
8
Time τ

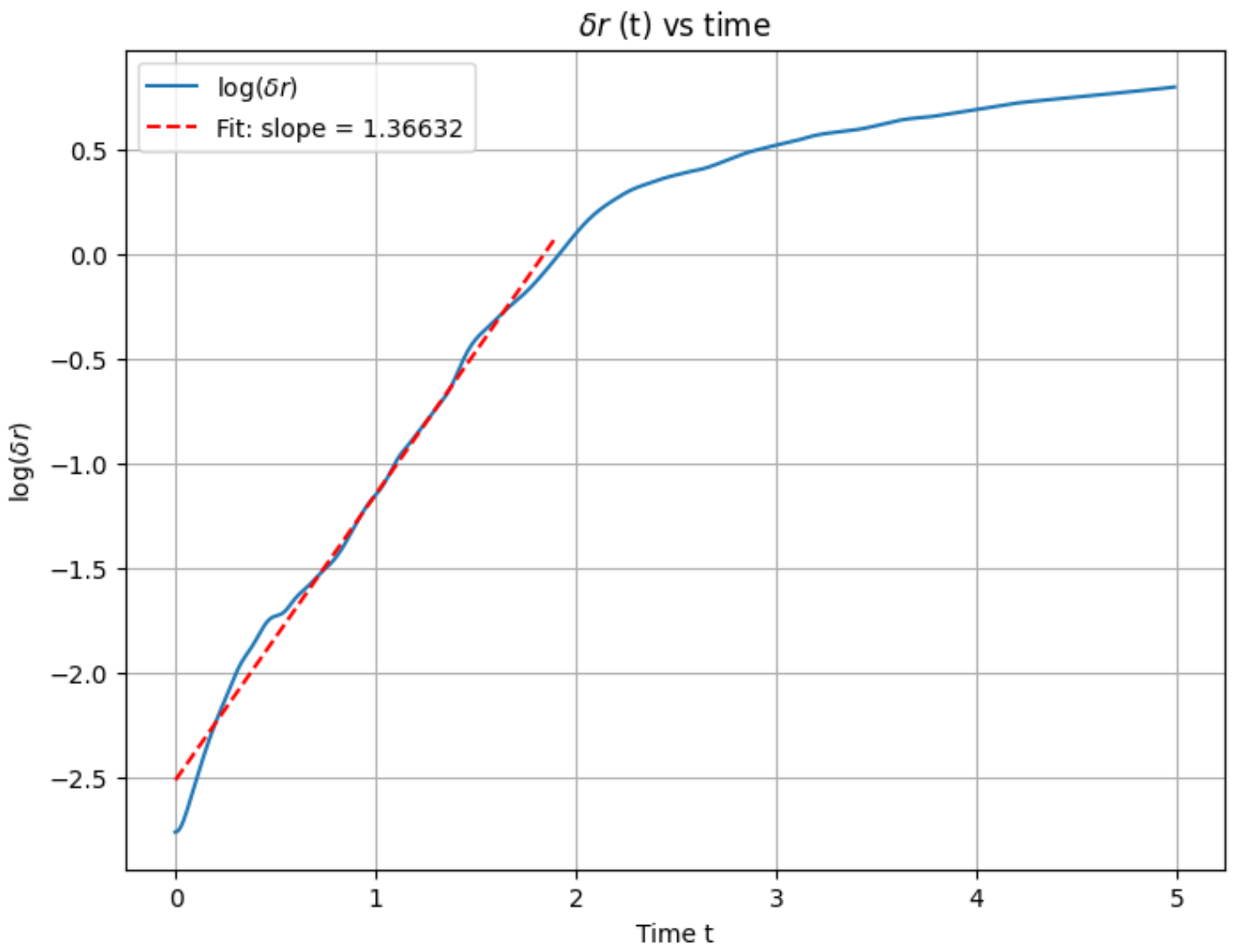
Calculation of Lyapunov Exponent Using 1000 Particles in Pairs

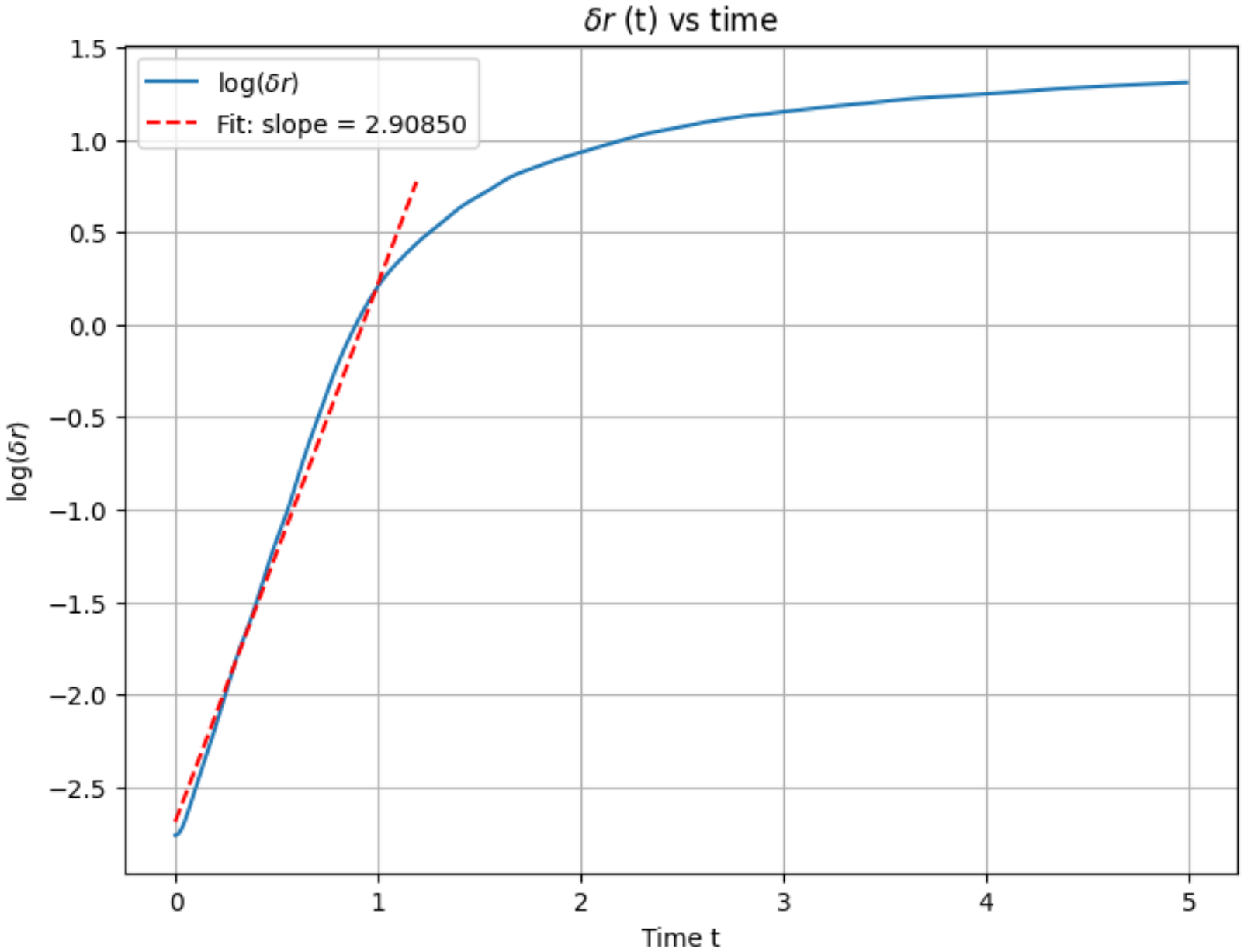
- To increase the statistical reliability of the Lyapunov exponent calculation, we consider a larger sample by using 1000 particles, organized into 500 pairs.
- For each pair, we calculate the difference in x, y, and θ at each time step:

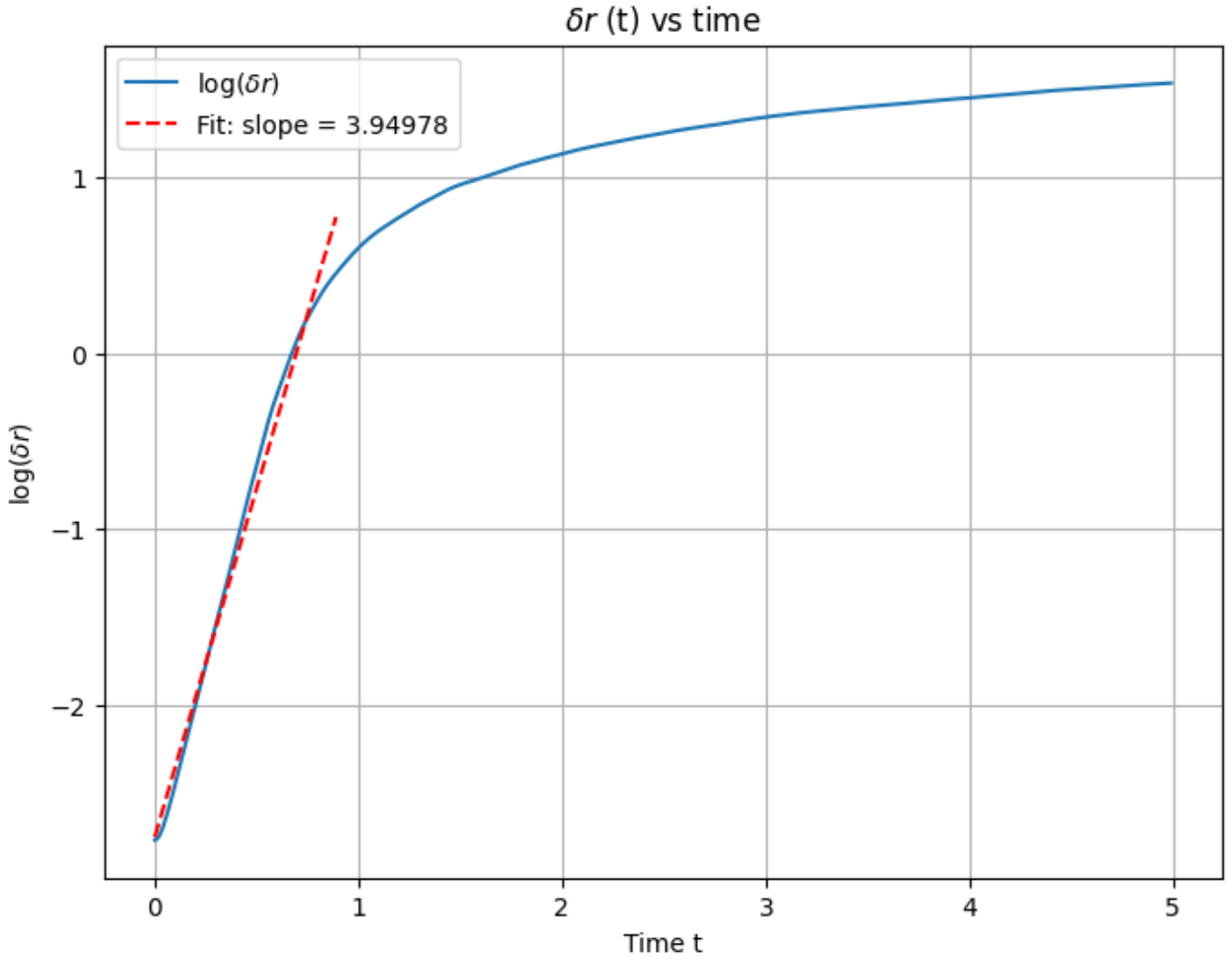
$$\delta x(t) = x_1(t) - x_2(t)$$
$$\delta y(t) = y_1(t) - y_2(t)$$
$$\delta \theta(t) = \theta_1(t) - \theta_2(t)$$

- We then calculate the Euclidean norm of these differences:

$$\delta r(t) = \sqrt{\delta x(t)^2 + \delta y(t)^2 + \delta \theta(t)^2}$$




δr (t) vs time
log(δr)
Fit: slope = 2.90850
log(δr)
Time t
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
0
1
2
3
4
5



"This trend clearly shows that higher v_s leads to a higher Lyapunov exponent, indicating stronger chaos in the system."

Concluding Remarks

- To summarize, our work explored the rich dynamics of active particles in structured fluid flows.
- Using MSD, we classified particle motion into different regimes: ballistic, diffusive, and confined. This analysis provided insight into how particle dynamics change with propulsion speed.
- To quantify chaos, we calculated Lyapunov exponents. We observed that higher v_s values led to larger exponents, confirming increased chaotic behavior.
- These results have applications in biology, technology, and environmental science, where understanding particle motion in complex flows is critical.
- Finally, our findings set the stage for future work, exploring more realistic systems.